

Star chromatic index of cubic graphs

OPG-37271

L1 - TRACTABLE

PARTIALLY SOLVED

Graph Theory

ORIGINAL PROBLEM STATEMENT

The star chromatic index $\chi'_s(G)$ of a graph G is the minimum number of colors needed to properly color the edges of the graph so that no path or cycle of length four is bi-colored.

Question Is it true that for every (sub)cubic graph G , we have $\chi'_s(G) \leq 6$?